Music Recommendation System Using Content Based Filtering

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Abstract

This paper presents the design and implementation of a content-based music recommendation system using machine learning techniques. The system processes song metadata, including genre, artist, and descriptive textual features, and transforms them into numerical representations using TF-IDF vectorization. Cosine similarity is then applied to measure similarity between songs and generate personalized recommendations based on a given input track. Experimental results show that the proposed approach produces meaningful recommendations by leveraging textual similarity. This work highlights the effectiveness of content-based filtering and the role of machine learning in improving music discovery and user experience.

I. Introduction

Recommendation systems now lie at the heart of most modern digital platforms, which allow for intelligent content discovery in various domains, from music and movies to shopping and social media. In the music industry alone, services like Spotify, YouTube Music, and Apple Music are now deeply dependent on Machine Learning algorithms to analyze vast amounts of data to provide highly personalized listening experiences.

The aim of this project is to construct a simplified version of such a system that will understand the characteristics of songs and user preferences, allowing it to generate meaningful recommendations. Focusing on genre information, mood-related attributes, and descriptive text, this system uses TF-IDF and cosine similarity to identify relations between songs and recommend those that best fit user preferences.

This report describes the background of recommendation systems, explains the methods applied in our system, and states our findings, limitations, and possible improvements that can be made in the future.

II. Background and Related Work

Music Recommendation Systems began as an answer to a problem brought by the increasing growth of digital music libraries when users have to deal with too much content and are struggling to find music that would correspond to their preferences. With the shift from physical media to online streaming, there arose a dire need for intelligent systems to organize, filter, and personalize the consumption of music. Early methods of recommendations depended on simple metadata such as artist name and genres, but soon with better data collection, user interaction tracking, and computing power, more advanced personalization techniques became possible.

First evolving through the addition of user behavior analysis, signal processing, and machine learning techniques, Music Recommendation Systems strove to understand both musical characteristics and the

preferences of listeners. Today, the systems analyze audio features, listening history, ratings, and contextual information to provide recommendations. They play an important role in enhancing user experience, increasing engagement, and supporting music discovery on Spotify, Apple Music, YouTube Music, and other online services, therefore being an integral part of modern digital music ecosystems.

Music recommendation systems are commonly classified into
three major categories:

1) Content-Based Filtering

This method recommends songs that are similar to those a user
prefers by analyzing song attributes such as genre, lyrics, and
descriptive tags. This is the primary approach adopted in this
project.

2) Collaborative Filtering

Collaborative filtering relies on user interaction data to identify similarities between users or items. Although
effective, it requires large-scale user data, and was outside the
scope of this study.

3) Hybrid Recommendation Systems

Hybrid systems combine content-based and collaborative filtering techniques to improve accuracy but require more complex data and computational resources.

III. Why do *we need a Music Recommendation System?*

There are millions of songs released every year, which makes it really hard to find new music on our own. Without help, users struggle to find music they like because there's too much to go through. This is where machine learning comes in - it acts like a smart assistant that learns what kind of music you enjoy. Machine learning helps us discover new favorite songs quickly without spending hours searching.

The problem got worse when music moved from CDs to online streaming. Now with platforms like Spotify and Apple Music, we have access to millions of songs. Without recommendation systems, finding good music would be like searching for a needle in a haystack.

IV. How Music Apps Learn Our Taste

Music apps like Spotify or Apple Music collect information about everything we do while listening to music. This information becomes data that the app can study.

What Data Gets Collected?

The app tracks what we listen to often - like if you play Taylor Swift songs five times a day. It watches what we skip immediately, which tells the app "I don't like this song." When we like or save songs, that's another clue about our taste. The app also pays attention to which artists and genres we follow. Even the time and mood of our listening matters - like calm music at night or energetic music in the morning.

What Happens to This Data?

Machine learning analyzes all this information to figure out what kind of music we'll probably enjoy next. The system looks at all these different pieces together to build a picture of what you like. For example, if you always skip country music but finish every pop song, the app learns to recommend more pop and less country.

V. What Machine Learning Learns

The ML system looks at four main things we do and learns from each one:

Play

When you finish listening to a song, that tells the system "I enjoyed this." The more you play certain types of songs, the more the system understands your preferences.

Skips

If you skip a song really fast, that's a strong message saying, "I don't like this." This helps the app avoid recommending similar songs. If you skip lots of heavy metal songs, it stops suggesting that genre.

Likes

When you press the like button, you're telling the app "I really love this style, artist, or mood." This is one of the strongest signals because you took the time to click something.

Playlist Additions

When you add songs to your playlists, it shows these are your favorites that you want to hear again and again. These are songs you really care about.

After studying all these actions, the system recommends songs similar to what you liked and avoids songs like the ones you skipped.

VI. How Machine Learning Recommends

The recommendation process works in four simple steps:

Step 1: Collect Listening Data

First, the app collects all the information about what you listen to, skip, like, and add to playlists. This is the foundation - without data, the system can't learn your preferences.

Step 2: Learn User Preference

Next, the ML system studies that data to figure out your music taste. It learns things like "this person likes pop music with upbeat tempo" or "this person likes sad indie songs."

Step 3: Find Similar Patterns

The system searches for patterns - it tries to find other songs that match what you like or finds other users who have similar tastes to you.

Step 4: Recommend Songs

Finally, the app gives you recommendations using two main methods:

Content-Based Filtering recommends songs similar to ones you already like. It analyzes song attributes like genre, artist, and descriptive tags. For example, if you love a Drake song, it recommends other Drake songs or similar music. The system uses **TF-IDF** (Term Frequency-Inverse Document Frequency) to combine song information like track name, artist, and genre into text descriptions, then uses **cosine similarity** to measure how similar songs are to each other.

Collaborative Filtering looks at what other people with similar taste enjoy. If you and another user both like 80% of the same songs, the app recommends songs that the other person likes but you haven't heard yet. It's like having friends with similar music taste - if they discover something good, you'll probably like it too.

Strengths and Weaknesses

What Works Well

Content-based filtering works even for new users because it just looks at the songs themselves, not other users' data. It's also good at explaining recommendations: "We recommended this because it's pop music by a female artist."

What Doesn't Work Well

1. Only Looks at Text, Not Sound: The system only reads genre and artist names - it doesn't actually listen

to the music. Two "pop" songs can sound completely different.

2. No Personalization: Everyone who searches for the same song gets identical recommendations, regardless of their personal taste.

3. Dataset Imbalance: If there are more pop songs than jazz songs in the database, the system mostly recommends pop. Popular genres get recommended more often.

4. Gets Outdated Quickly: New songs come out daily, but someone has to manually add them to the database, which is impossible to keep up with.

5. No Audio Analysis: The system doesn't understand tempo, rhythm, or mood - just text descriptions.

VII. Dataset and Feature Description

The proposed system utilizes the **TCC CEDs Music Dataset**,
which contains metadata for a wide range of songs across
multiple decades. Key attributes include track name, artist
name, genre, popularity, duration, and release information.
These features serve as the basis for computing textual
similarity between songs.
Although the dataset covers a diverse range of music, it
exhibits class imbalance, with certain genres and artists
appearing more frequently. This imbalance influences recommendation results by favoring dominant categories.

VIII. Preprocessing and Feature Engineering

Effective preprocessing was essential for accurate model
performance. The applied steps include:

- **Lowercasing:** All text features were converted to lowercase to ensure consistency.
- **Cleaning:** Removal of special characters, extra spaces, and irregular formatting.
- **Duplicate Removal:** Eliminating redundant entries to avoid biased recommendations.
- **Feature Merging:** Combining track name, artist name, and genre into a single textual feature.

These steps resulted in a standardized and clean dataset
suitable for TF-IDF vectorization.

IX. System Architecture and Model Design

The system architecture follows a sequential pipeline:
Input → Text Preprocessing → TF-IDF Vectorization →
Cosine Similarity → Recommended Songs
TF-IDF assigns weights to terms based on their importance
within the dataset, while cosine similarity measures the
closeness between song vectors in multidimensional space.

X.Technical Implementation

The system was implemented using Python with the following
libraries:

- **Pandas:** Data loading and preprocessing
- **NumPy:** Numerical operations
- **Scikit-learn:** TF-IDF vectorization and cosine similarity computation

Given an input song, the system computes similarity scores
against all other songs and outputs the top recommended
tracks.

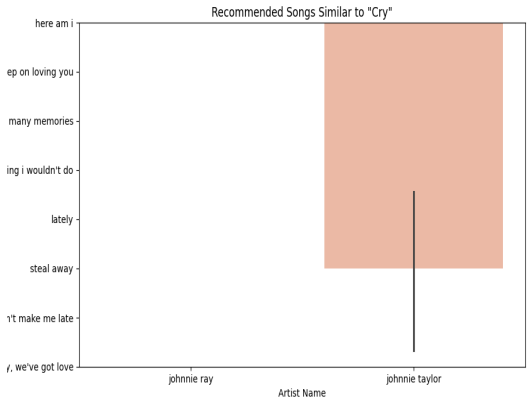
XI. Results

The discussed music recommendation system produced a set of songs that were *similar* to the input provided by the user (i.e, it uses a content-based recommendation system) using the metadata such as track name, artist name, music genre, etc. When a seed song was provided, in this case “Cry”, the model generated the top recommendation tracks by computing cosine similarity between TF - IDF (Term Frequency - Inverse Document Frequency) vectorised metadata. A list of songs along with their corresponding artists and genre information, which were *assumed* to be similar to the input song, was printed out as output, indicating the closest matches according to textual similarity.[1]

The system also generated a visualization image to support the results, as you can see in Figure 1. A bar chart was produced in which the recommended songs were displayed along the y-axis and their respective artists along the x-axis. [1]

Before generating the recommendations, the dataset was examined using exploratory data analysis (EDA). This step gave weight and values to each feature of the song and showed how songs were distributed across different genres and artists, and also confirmed that the dataset used for this system consisted of music from a range of 100 years.[1]

As observed from Fig. 2 and Fig. 3, the visual graphs show the top 10 genres and artists dominating the music industry based upon the provided dataset. This is important because if certain genres and artists are more infamous



than others, it is highly likely that this music recommendation system will recommend songs from those categories more often.[1]

Although it seemingly seems that the system recommends *similar* songs to the user, however as it is just a content-based recommendation, and it does not take user behaviour into account, which is its greatest weakness. Hence, the original authors note that the model remains a preliminary prototype and cannot be used as an official model to recommend songs in the market.[3]

Fig 1: Recommended songs

XII. DISCUSSIONS

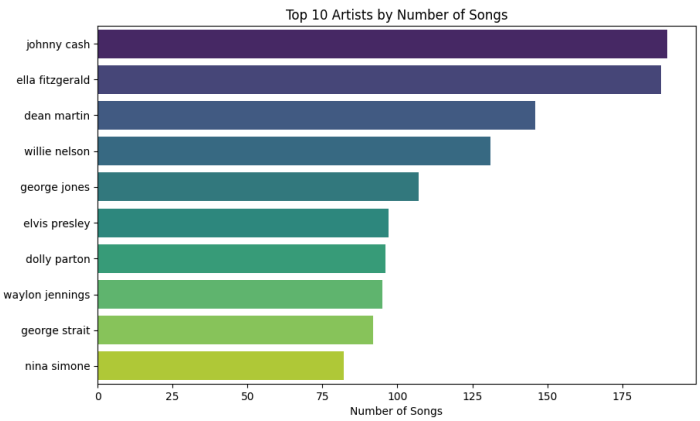


Fig 2: Top artists

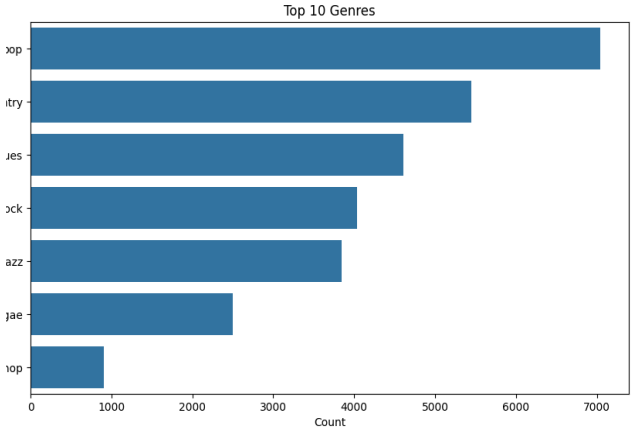


Fig 3: Top genre

While the system was able to recommend songs similar to the input based on the metadata provided, it has several limitations. Its greatest weakness is that it is solely dependent only on a content-based method, meaning the model only considers textual information such as artist names, genres, track, title, certain words, etc but music is so much more than that. It ignores the deeper musical qualities such as tempo, rhythm or mood. As a result, the recommendations may be similar in description but not necessarily in how they actually sound.

Another major weakness is the lack of personalization. The system does not use any user behaviour or listening history, meaning every user will receive the same recommendations for the same input song. This makes the system far less effective compared to real music platforms that rely heavily on user preferences.

The dataset used for this system is also noticeably imbalanced, with certain genres and artists appearing much more frequently than others. As shown in the EDA figures, this causes the model to recommend songs mainly from these dominant categories, which limits the variety of suggestions provided to the user. This problem becomes even more significant when considering that the dataset must be manually updated. With new songs being released constantly, updating the dataset by hand becomes practically impossible, meaning the system will quickly fall behind and fail to include newer music.

Given all these limitations, this model can really only be seen as a simple example of how content based filtering works. It still needs a lot more development, like including actual audio features, understanding user behaviour, and having the dataset update on its own, before it can properly work as a real music recommendation system.

XIII. Challenges and Limitations

- Lack of user personalization
- Dependence on metadata rather than audio features
- Dataset imbalance
- Manual dataset updates
- Limited scalability

XIV. Conclusion

This project shows how content-based filtering using TF-IDF and cosine similarity can effectively recommend music in a natural and user-friendly way. By examining song features such as genre and descriptive information, the system identifies similarities between tracks and delivers recommendations that closely match individual preferences. As a result, users are able to discover new music that aligns with their taste without feeling overwhelmed by too many choices.

The findings demonstrate how machine learning plays an important role in improving digital music platforms by making discovery more personalized and engaging. Overall, this project emphasizes that personalization is a key factor in successful recommendation systems, as it enhances user satisfaction and creates a more enjoyable listening experience across modern music applications.

XV. Future Work

Future Enhancements include:

1. **Employing Techniques of Collaborative Filtering**

In the future, collaborative filtering will be integrated to analyze user–user and item–item interactions. The system can learn from patterns of listens, ratings, and preferences of similar users to provide more diverse and personalized recommendations that extend beyond the individual content similarities.

2. **Extracting Audio Features Using Libraries such as Librosa**

Low-level and high-level features of the audio can be extracted, like tempo, MFCCs, spectral contrast, and rhythm, using the Librosa library, thus enhancing the system. Such features permit insight into musical characteristics in depth and allow the calculation of similarity between songs with higher accuracy.

3. **Developing a Hybrid Recommendation Model**

A hybrid model will be used to combine content-based with collaborative filtering techniques.

This approach mitigates the limitations found in the cold-start problem and sparsity, ensuring that recommendations are well-balanced between song attributes and user behaviors.

4. **Implementing a User Interface for Real-Time Recommendations**

The interactive user interface will introduce users to the real-time functionality of receiving music recommendations. Users will be able to search and select songs and immediately see the recommended tracks; this usability and engagement is achieved through seamless and friendly user experiences.

5. **Dataset Updates: Expansion and Automation**

The dataset will be enhanced by increasing the size of the music data and enabling the updates through data pipelines or APIs. Regular updates ensure that newly released songs and evolving user preferences are captured, keeping recommendations relevant and up to date. These improvements will vastly enhance the system's scalability, accuracy, and user experience, making it more robust and adaptable for real-world music streaming platforms.

References

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